Smartphone-Based Indoor Localization System Using Wi-Fi RTT/Magnetic/PDR Based on an Improved Particle Filter

Meng Sun, Yunjia Wang, Nanshan Zheng, Guoliang Chen, Zengke Li, Jingxue Bi, and Hongchao Yang

***Abstract*— In this article, we have constructed a high-precision indoor positioning system by leveraging Wi-Fi round-trip time (RTT), magnetic field, and pedestrian dead reckoning (PDR). For the Wi-Fi RTT module, a collaborative Wi-Fi RTT positioning approach is proposed by combining RTT ranging and fingerprinting through a Cramer–Rao lower bound (CRLB)-based outlier detection method. Based on this, an enhanced version of the geomagnetic-aided Wi-Fi RTT (GAWR) localization method is implemented. For better smartphone-based deployment, we improve PDR in terms of step detection, step length estimation, and heading optimization. Moreover, an optimization strategy incorporating map-aided resampling and particle swarm reconstruction is designed to enhance the performance of particle filter (PF). Finally, the Wi-Fi RTT, magnetic positioning, and PDR are integrated based on the improved PF (IPF). Our system employs two mechanisms to achieve fusion localization, which means that the system executes Wi-Fi/magnetic/PDR fusion when the Wi-Fi module is available and switches to magnetic/PDR fusion when the Wi-Fi signal is absent. Extensive experiments conducted in a corridor environment demonstrate that the IPF-based Wi-Fi RTT/magnetic/PDR model achieves the localization accuracy of 0.64–0.91 m when the Wi-Fi module is available and unavailable, respectively, showing meter-level continuous positioning capability even in the presence of additional Wi-Fi ranging errors and in areas without Wi-Fi signal coverage.**

***Index Terms*— Fingerprinting, indoor localization, magnetic field, particle filter (PF), pedestrian dead reckoning (PDR), round-trip time (RTT), sensors fusion, Wi-Fi fine time measurement (FTM).**

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I. Introduction

INDOOR localization has emerged as a prominent research area within the positioning and navigation community. To estimate indoor positions, a diverse range of wireless signals have been employed, such as Wi-Fi [1] and ultrawideband [2]. With the popularization of smartphones, performing indoor positioning on mobile devices has become a significant research direction. Leveraging integrated sensor modules, smartphone-based techniques have undergone extensive study. As the wireless communication module of mobile phones, 5G positioning holds the potential to deliver ubiquitous and high-precision indoor localization services [3], while its practical application is limited by the sparse deployment of base stations in real-world scenarios.

The recently released Wi-Fi fine time measurement (FTM) approach [4] enables Wi-Fi ranging positioning through round-trip time (RTT) measurement. Researchers have conducted works from perspectives, including Wi-Fi ranging influencing factors [5], ranging compensation model [6], nonlight-of-sight (NLoS) error identification [7], positioning optimization [8], FTM fingerprinting [9], and fusion model construction [10], [11], [12]. Generally, meter-level Wi-Fi ranging can be reached with appropriate compensation in line-of-sight (LoS) conditions, as reported in [5]. More advanced ranging compensation models such as artificial neural networks [6] and recurrent neural networks [13] can further enhance ranging precision. NLoS identification aids in improving localization accuracy by mitigating the NLoS errors in Wi-Fi ranging [7]. However, these efforts still struggle to address the instability of Wi-Fi ranging positioning caused by the complex indoor topologies, prompting researchers to explore FTM fingerprinting [9] or integrate Wi-Fi RTT with pedestrian dead reckoning (PDR) [11]. With the integration of PDR, both the accuracy and stability of Wi-Fi RTT positioning are better improved.

Apart from the well-established wireless positioning methods, there has been significant discussion on infrastructure-free localization techniques. Among these, PDR [10] and magnetic positioning [14] stand out as the most popular approaches. PDR is a kind of relative positioning method that uses accelerators, gyroscopes, and magnetometers to offer continuous location outputs without the need for additional infrastructure.

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However, it suffers from the challenge of error accumulation. To address this issue, PDR optimization efforts have been directed at heading estimation [15], step length estimation [16], and step counting [17]. In addition, research into walking pattern identification [16] has contributed to enhancing the robustness of PDR. Despite extensive work that has been carried out, it is still a great challenge to achieve stable and reliable PDR on mobile devices due to the inherent uncertainty of pedestrian activities. Generally, PDR is always integrated with other technologies to overcome error accumulation.

Geomagnetic positioning determines locations based on the spatially discernible indoor magnetic field. The pervasiveness of the magnetic field theoretically enables magnetic positioning to cover all indoor environments. By concluding the previous works, geomagnetic positioning can be achieved by using single-point-based magnetic positioning (SPMP) and sequence-based magnetic positioning (SBMP) [18]. SBMP uses sequential magnetic features, incorporating algorithms such as dynamic time wrapping (DTW) [19] and Gauss–Newton iterative (GNI) [20]. SPMP utilizes discrete features, often employing localization methods such as k-nearest neighbors (KNNs) [21] and stochastic magnetic matching [22]. SBMP boasts higher accuracy than SPMP due to the specificity of sequence features, while SPMP offers flexibility by not requiring magnetic sequence collection along a path. To improve positioning accuracy and robustness, geomagnetic positioning is commonly integrated with PDR to build an infrastructure-free fusion system, leveraging tools such as the extended Kalman filter (EKF) [23] and particle filter (PF) [24]. Testing results have confirmed that magnetic/PDR fusion can deliver promising ubiquitous positioning results. However, disturbances in the magnetic field arise challenges to accurate magnetic matching and the robustness of magnetic/PDR fusion systems.

Although numerous indoor localization methods have been developed, achieving low cost, high precision, robust, and ubiquitous indoor localization remains a significant challenge in complex indoor environments. For instance, Wi-Fi/PDR positioning systems can achieve high-precision localization within signal coverage areas, but they are unable to function effectively in areas with weak or no Wi-Fi signals. Deploying more Wi-Fi devices to ensure signal coverage results in higher positioning costs. Magnetic/PDR fusion is infrastructure-free, but magnetic data fluctuations might affect the system's stability. Therefore, the integration of Wi-Fi RTT, magnetic, and PDR holds the potential to bridge gaps in scenarios where the fusion of two methods may fail to deliver localization results. Our research focuses on the problems of system accuracy, complexity, and continuous positioning capability in the absence of Wi-Fi signals. The following issues are studied.

1) How to dynamically evaluate Wi-Fi RTT positioning and design collaborative RTT localization method.
2) How to improve the smartphone-based PDR framework for better deployment and error accumulation control.
3) How to effectively integrate Wi-Fi RTT/magnetic/PDR data to achieve high-precision and continuous localization without more power consumption.

Based on the abovementioned issues, we have proposed different strategies to address them. In our efforts to optimize RTT positioning, we employ the Cramer–Rao lower bound (CRLB) to dynamically evaluate online positioning results and use RTT fingerprinting to fill in the absence of detected outliers. Based on this, we introduce an enhanced version of the geomagnetic-aided Wi-Fi RTT (GAWR) positioning [25], which incorporates magnetic data to further improve Wi-Fi positioning accuracy. We analyzed the relationships between acceleration data and the walking process and proposed novel step detection and step length estimation models. The heading angle is optimized by using the strong correlation between the geomagnetic sequence and the walking path. We also designed strategies to improve PF and developed an improved PF (IPF)-based Wi-Fi RTT/magnetic/PDR fusion system. Our contributions are summarized as follows.

1) We propose a CRLB-based Wi-Fi RTT collaborative positioning approach, which incorporates RTT ranging and fingerprinting methods by using an online CRLB-based outlier detection strategy. Building upon this foundation, an enhanced GAWR (EGAWR) positioning approach is presented. Testing results demonstrate that the proposed methods improve the accuracy and stability of RTT positioning.
2) We improve the PDR framework from the aspects of taking steps, walking distance, and heading estimation. The improved PDR approach incorporates a binarization-based step detection algorithm, an automatic step length estimation model, and a geomagnetic waveform-based heading optimization method. All the proposed methods are user-friendly on smartphones and outperform the performance of the compared models.
3) We develop a Wi-Fi RTT/magnetic/PDR integration system by using an IPF, which is optimized through map-aided resampling and particle swarm reconstruction strategies. Compared with state-of-the-art fusion methods, our system shows advantages in positioning precision and continuous localization capability when transitioning between areas with and without Wi-Fi signal coverage.

The rest of this article is organized as follows. The related works are reviewed in Section II. The system architecture is introduced in Section III. Section IV presents the proposed methods and Section V showcases the testing results and makes discussions. Section VI concludes the work.

II. Related Works

A. Wi-Fi RTT Localization

Since the Andoird system supports the Wi-Fi FTM protocol, smartphone-based Wi-Fi FTM positioning has been attracting significant attention. Ibrahim et al. [5] analyzed the influencing factors of Wi-Fi RTT ranging and concluded that meter-level Wi-Fi ranging can be achieved in LoS conditions.

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An artificial neural network [6] is adopted for RTT ranging compensation via supervised deep learning. The ranging errors within 1 m are in 60% of the cases. In [13], the recurrent neural networks model achieves an average ranging accuracy of 0.5 m. To improve ranging accuracy, researchers utilized regression methods such as nonlinear least squares (LS) [26] to compensate for the ranging data. Wi-Fi RSS has also been employed for RTT ranging accuracy improvement [27], where real-time measured RTT data are combined with transformed RSS-based distance to enhance RTT ranging accuracy.

Wi-Fi RTT positioning optimizations have been made from various aspects. For instance, Si et al. [7] used the Gaussian model to distinguish and eliminate ranging data with NLoS errors, thereby improving localization accuracy. Banin et al. [8] proposed a map-aided FTM positioning method and obtained a positioning accuracy within 3 m in 90% of the cases. The simultaneously measured Wi-Fi RSS provides favorable conditions for FTM fingerprinting. Based on the RTT and RSS data, Huilla et al. [9] explored FTM fingerprinting using weighted KNNs (WKNNs) and obtained an accuracy of 2.13 m in 95% of the cases. Eberechukwu et al. [28] proposed a DNN-based positioning model, and obtained results of 0.66–1.5 m for the 2σ percentile errors in two datasets. Rizk et al. [29] proposed an *RRLoc* FTM fingerprinting system that obtains an average positioning accuracy of 0.51–0.59 m in the office and laboratory testbeds. Martin-Escalona and Zola [30] utilized different classification algorithms (e.g., WKNN and SVM) to investigate RTT and RSS-based fingerprinting methods. Testing results reveal comparable performance between RSS fingerprinting and RTT ranging positioning.

B. Geomagnetic Positioning

Generally, because of the high-dimensional data, magnetic sequence features are suitable for pattern identification or machine learning. In [14], the probabilistic neural network is built to perform magnetic matching, and the highest correct matching rates of 94%–100% in two regions are obtained. The DTW method is also employed to recognize the magnetic patterns of different hallways [19], achieving a high-precision recognition rate. In [20], GNI is designed to find the optimally aligned reference magnetic sequence of the database. Testing results show that combining GNI with PDR delivers a mean accuracy of 2.34 m. Moreover, convolutional neural network [31] is introduced to the "MINLOC" system, which builds the connection between the magnetic patterns and actual positions, achieving accuracy within 1.01 m in 75% of the cases.

Regarding the matching methods of SPMP, the KNN [21] approach is frequently used for the similarities assessment of the magnetic features. However, due to the low-dimensional data (normally three-axis components of the magnetic field), the positioning performance is not good. To solve this problem, an orientation-aided stochastic magnetic matching is proposed in [22], which introduces orientation to form the magnetic fingerprints and match the crowdsourced magnetic data using histograms, delivering location errors of 2.4 m. In [32], a multimagnetic-fingerprint-fusion is proposed, which combines coarse estimation with fine estimation to improve magnetic positioning accuracy. The results show that 99% cumulative distribution function (cdf) of location error is less than 0.5 m. In [33], the "*ILoA*" system that uses the direction-variant augmented vector of the magnetic field is presented. *ILoA* employs PFs and obtains promising results.

C. PDR Positioning

PDR contains modules of step detection, heading estimation, and step length estimation. The heading estimation methods, including the gradient descent method [15], heading compensation [34], and adaptive EKF (AEKF) [35], have been thoroughly studied. For instance, the heading angle problem with different tilt angles of phones is addressed in [34]. The direction cosine matrix (DCM) associated with pitch angles is defined and compensated for the DCM that transforms vectors from the navigation frame to the body frame. In [35], the AEKF is employed to smooth the sensor data and the heading drift is about 4°. Regarding step detection methods, researchers have studied the relationship between the process of taking steps and accelerations. Some of these frequently used algorithms include the finite-state machine (FSM) [36], peak detection [37], and the zero velocity update method [38]. In [36], one-step period is divided into five states by FSM and the process of taking steps is identified if all these conditions are satisfied. The peak detection approach [37] finds the peaks and valleys of acceleration to detect steps. The shortcomings of these methods are that they are dependent on the acceleration curves and sensitive to false peaks of accelerations, which exhibit a similar process to a normal step.

As for the step length estimation models, researchers have proposed various approaches, including the machine learning model [16], the linear model [39], and the nonlinear model [40]. Machine learning methods outperform the linear and nonlinear models in terms of accuracy, while the latter two methods offer easy operation on mobile phones with low time complexity.

D. Fusion Localization

The integration styles of Wi-Fi, magnetic, and PDR are commonly combinations of two methods, such as magnetic/PDR [23] and Wi-Fi RTT/PDR [36]. Filter algorithms are usually adopted into Wi-Fi RTT/PDR fusion. Guo et al. [10] introduced a robust fusion framework that combines Wi-Fi RTT, RSS, and PDR using an adaptive Kalman filter (KF). Wi-Fi ranging compensation and observation quality evaluation and control strategy are investigated to improve the accuracy and robustness of the system, and the average positioning accuracy is 0.572 m. Zhou et al. [11] constructed a Wi-Fi RTT/encoder/INS-based fusion model based on an AEKF. The system reduced the location errors of Wi-Fi RTT positioning by 54.62%–58.38% under LoS and NLoS conditions. Xu et al. [34] integrated PDR with Wi-Fi RTT by using an enhanced PF, which is improved using map information. The localization accuracy of the system is within 1 m in 86.7% of the cases.

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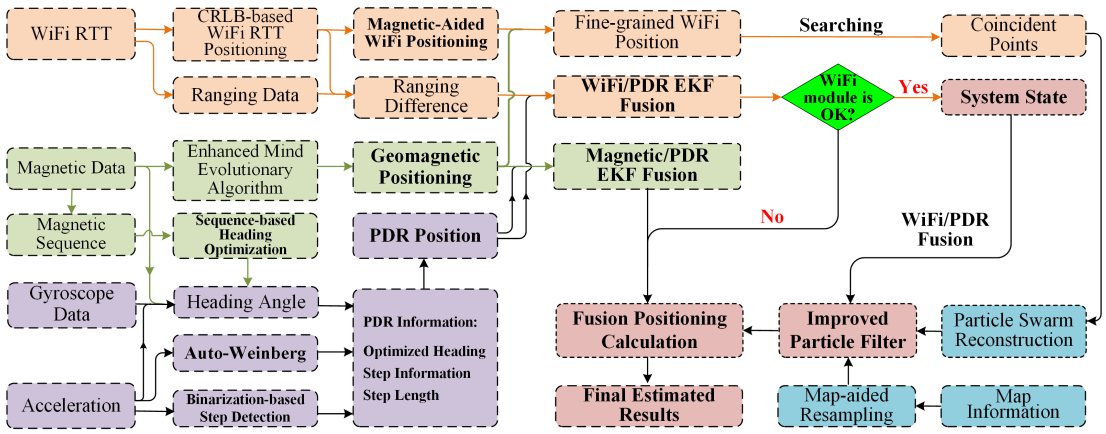

Fig. 1. System architecture of Wi-Fi RTT/magnetic/PDR fusion.

Filter algorithms are often applied in magnetic/PDR fusion. In [23], the magnetic positioning results are integrated with PDR based on the EKF, achieving a mean location error of about 1.2 m in the testing site. In [24], a Wi-Fi/magnetic/PDR fusion model is presented based on an IPF. The magnetic matching is restricted in a determined area using Wi-Fi, and the magnetic positioning is combined with PDR, delivering an accuracy of 1 m in 80% of the cases. However, the Wi-Fi module in [24] is not the Wi-Fi RTT technique, but just the traditional Wi-Fi fingerprinting. In [41], a trajectory-guided 2-D magnetic map construction is proposed to improve the accuracy of magnetic positioning, an IPF is presented to deal with the magnetic/PDR fusion problem, and an average positioning error of 0.32 m is obtained.

III. System Architecture

Fig. 1 shows the flowchart of the proposed Wi-Fi RTT/magnetic/PDR fusion system, which relies on an IPF, optimized using both map information (for improving resampling) and Wi-Fi RTT positioning (for improving particle diversity). The Wi-Fi module dynamically evaluates the ranging positioning results through the CRLB-based outlier detection (Section IV-A). Based on this, we achieve an enhanced version of the GAWR positioning method (Section IV-B), which improves the localization accuracy of magnetic and Wi-Fi positioning. (Section IV-B). The PDR module estimates walking direction and distance, furnishing information for particle propagation. The magnetic positioning module utilizes an enhanced mind evolutionary algorithm (EMEA) to obtain accurate results, as proven in our work [23].

After every module is initialized, the system employs the IPF to integrate Wi-Fi/magnetic/PDR when the Wi-Fi RTT module is accessible. In this case, the system kinematic model is defined by the EKF-based Wi-Fi/PDR tightly fusion [36]. The final fusion result is generated by computing the average coordinates of the resampled particles. In another case where the Wi-Fi module is unavailable, the fusion system transitions to the EKF-based magnetic/PDR fusion [23] to ensure continuity. We will present the details of the abovementioned positioning modules in Section IV.

IV. Methods

A. CRLB-Based Wi-Fi RTT Positioning

1) CRLB Derivation of Wi-Fi RTT-Based Positioning: The CRLB offers an efficient method to evaluate the performance of an unbiased estimator [42]. Considering a localization scheme with n Wi-Fi APs, denoted as $\boldsymbol{s}_i = [x_i, y_i]^T \in \mathbb{R}^2, i \in \{1, 2, \ldots, n\}$, the target $\boldsymbol{t} = [x, y]^T$, given the measured RTT ranging data $\boldsymbol{d}$, the location problem involves finding the target estimation $\boldsymbol{t} = [x, y]^T$ using $\boldsymbol{s}$ and $\boldsymbol{d}$. The estimation variance of $\hat{\boldsymbol{t}} = [\hat{x}, \hat{y}]^T$ is bounded by the CRLB. The measured RTT data is modeled as

$$\hat{d}_i = d_i + \zeta, \quad \zeta \sim \mathcal{N}\left(0, \sigma_i^2\right) \tag{1}$$

where d_i and $\hat{d}_i$ are the theoretical and measured distances between the target and the ith Wi-Fi AP, respectively; $d_i = ||\boldsymbol{t} - \boldsymbol{s}_i||_2$; and ζ describes the ranging error following a Gaussian distribution with zero mean and variance σ_i^2. It should be noted that different APs correspond to different σ_i^2's during practical measurements. Assuming that the RTT observations from different APs are independent, the joint probability density function (pdf) is defined as

$$f\left(\hat{\boldsymbol{d}}|\boldsymbol{t}\right) = f\left(\hat{d}_1|\boldsymbol{t}\right) \times f\left(\hat{d}_2|\boldsymbol{t}\right) \times \cdots \times f\left(\hat{d}_n|\boldsymbol{t}\right) = \prod_{i=1}^{n} f\left(\hat{d}_i|\boldsymbol{t}\right) \tag{2}$$

where $\hat{\boldsymbol{d}}$ represents the simultaneously captured RTT ranging data from n APs and $\hat{\boldsymbol{d}} = \{\hat{d}_1, \hat{d}_2, \ldots, \hat{d}_n\}^T$, $\boldsymbol{t}$ denotes the true position of the target. $\boldsymbol{t}$ can be estimated using maximum likelihood estimation as follows:

$$\boldsymbol{t} = \arg\max \ln f(\hat{\boldsymbol{d}}|\boldsymbol{t}). \tag{3}$$

Applying logarithmic operations, we obtain

$$\begin{aligned} lnf\left(\hat{\boldsymbol{d}}|\boldsymbol{t}\right) &= \ln f\left(\hat{d}_1|\boldsymbol{t}\right) + \cdots + f\left(\hat{d}_n|\boldsymbol{t}\right) = \sum_{i=1}^{n} f\left(\hat{d}_i|\boldsymbol{t}\right) \\ &= M - \frac{1}{2}\left[\left(\hat{\boldsymbol{d}} - \boldsymbol{d}(\boldsymbol{t})^T\right)\boldsymbol{R}^{-1}(\hat{\boldsymbol{d}} - \boldsymbol{d}(\boldsymbol{t})\right] \end{aligned} \tag{4}$$

where $\boldsymbol{d}(\boldsymbol{t})$ represents the true theoretical distance between the target position and Wi-Fi APs, $\boldsymbol{R}$ is the measurement variance matrix concerning n Wi-Fi APs, and

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$\boldsymbol{R} = \mathrm{diag}\{\sigma_1^2, \sigma_2^2, \ldots, \sigma_n^2\}$. Based on the pdf of Gaussian distribution in (1) and the logarithmic operations in (4), M is defined as

$$M = \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi}\sigma_i}\right). \tag{5}$$

If the estimate covariance matrix of $\hat{\boldsymbol{t}}$ is $\boldsymbol{P}$, $\boldsymbol{P}$ is bounded by the inverse of the fisher information matrix (FIM) $\boldsymbol{I}$ as

$$\boldsymbol{P} = E\left\{\left(\boldsymbol{t} - \hat{\boldsymbol{t}}\right)\left(\boldsymbol{t} - \hat{\boldsymbol{t}}\right)^T\right\} \geq \boldsymbol{I}^{-1}(\boldsymbol{t}) \tag{6}$$

where $\boldsymbol{I}$ is nonsingular and further given by

$$\begin{aligned}\boldsymbol{I} &= E\left[\nabla \ln f\left(\hat{\boldsymbol{d}}|\boldsymbol{t}\right)\nabla \ln f\left(\hat{\boldsymbol{d}}|\boldsymbol{t}\right)^T\right] \\ &= -E\left[\nabla^2 \ln f\left(\hat{\boldsymbol{d}}|\boldsymbol{t}\right)\right] = \begin{bmatrix} I_{xx} & I_{xy} \\ I_{yx} & I_{yy} \end{bmatrix}\end{aligned} \tag{7}$$

where ∇ and ∇^2 denote the operator of first- and second-order differentiation, respectively. According to (4), we obtain the first-operator differentiation as

$$\nabla \ln f\left(\hat{\boldsymbol{d}}|\boldsymbol{t}\right) = \nabla \boldsymbol{d}^T(\boldsymbol{t})\boldsymbol{R}^{-1}\left(\hat{\boldsymbol{d}} - \boldsymbol{d}(\boldsymbol{t})\right). \tag{8}$$

Substituting (8) into (7), $\boldsymbol{I}$ is defined as follows:

$$\boldsymbol{I}(\boldsymbol{t}) = \nabla \boldsymbol{d}^T(\boldsymbol{t})\boldsymbol{R}^{-1}\nabla \boldsymbol{d}(\boldsymbol{t}). \tag{9}$$

The calculation method of the theoretical distance is as

$$d_i = ||\boldsymbol{t} - \boldsymbol{s}_i|| = \sqrt{(x - x_i)^2 + (y - y_i)^2}. \tag{10}$$

Therefore, the entries of (7) can be determined by taking differentiations with respect to the target position coordinates as

$$I_{xx} = \sum_{i=1}^{n} \frac{1}{\sigma_i^2} \frac{(x - x_i)^2}{\left(\sqrt{(x - x_i)^2 + (y - y_i)^2}\right)^2} \tag{11}$$

$$I_{yy} = \sum_{i=1}^{n} \frac{1}{\sigma_i^2} \frac{(y - y_i)^2}{\left(\sqrt{(x - x_i)^2 + (y - y_i)^2}\right)^2} \tag{12}$$

$$I_{xy} = I_{yx} = \sum_{i=1}^{n} \frac{1}{\sigma_i^2} \frac{(x - x_i)(y - y_i)}{\left(\sqrt{(x - x_i)^2 + (y - y_i)^2}\right)^2}. \tag{13}$$

The CRLB of RTT-based positioning is defined as

$$\sigma_C^2(\boldsymbol{t}) = \sigma_C^2(x) + \sigma_C^2(y) = \mathrm{tr}(\boldsymbol{I}^{-1}(\boldsymbol{t})) \tag{14}$$

$$\mathrm{tr}(\boldsymbol{I}^{-1}(\boldsymbol{t})) = \frac{\mathrm{tr}(\boldsymbol{I}(\boldsymbol{t}))}{\det(\boldsymbol{I}(\boldsymbol{t}))} = \frac{I_{xx} + I_{yy}}{I_{xx} \times I_{yy} - I_{xy} \times I_{yx}} \tag{15}$$

where $\mathrm{tr}\{\bullet\}$ and $\det\{\bullet\}$ mean taking the trace and determinant of the matrix, respectively. Equation (14) reveals that the CRLB of RTT-based positioning describes the summation of the CRLBs of the x- and y-coordinates.

From (11) to (15), it can be concluded that the calculation of CRLB depends on the positions of Wi-Fi APs and the target position, the number of APs, and the measurement variances corresponding to different APs. As illustrated in Fig. 2, we observe that the relationship between the variances of RTT ranging and the ground-truth distance can be described as a linear trend. Therefore, we model this pattern as follows:

$$\boldsymbol{\sigma} = \boldsymbol{k}\boldsymbol{d} + \boldsymbol{b} \tag{16}$$

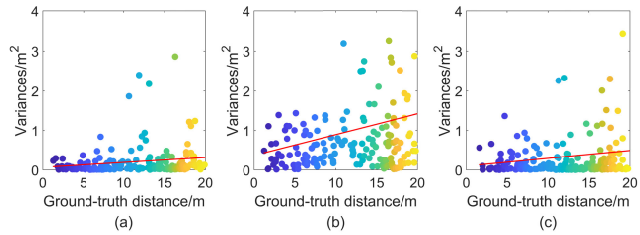

Fig. 2. Variances of ranging errors of different Wi-Fi APs. (a) No. 1 AP. (b) No. 2 AP. (c) No. 3 AP.

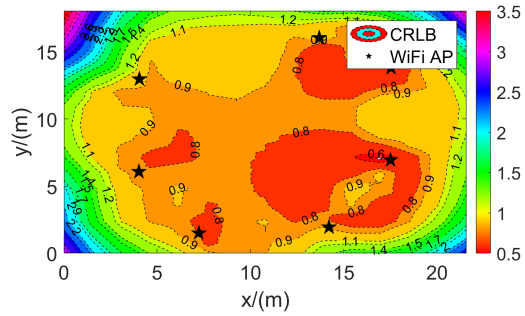

Fig. 3. CRLB distribution of the testing area of this work.

where $\boldsymbol{d}$ is the matrix of distances, $\boldsymbol{\sigma}$ is the variance matrix; $\boldsymbol{\sigma} = [\sigma_1, \ldots, \sigma_n]^T$; and $\boldsymbol{k}$ and $\boldsymbol{b}$ represent the coefficient matrices, $\boldsymbol{k} = \mathrm{diag}\{k_1, \ldots, k_n\}^T$ and $\boldsymbol{b} = [b_1, \ldots, b_n]^T$, respectively.

Utilizing the predicted measurement variances alongside the coordinates of Wi-Fi APs and testing points, the CRLB values at the testing points can be calculated. Then, using a linear interpolation algorithm, we construct the CRLB map of the testing area (as shown in Fig. 3). The distribution of CRLB indicates that the further away from the Wi-Fi AP, the greater the CRLB values. This observation serves as valuable insight for evaluating the credibility of the RTT positioning results and enabling us to dynamically identify and remove outliers.

2) CRLB-Based Collaborative RTT Positioning Approach: Fig. 4 shows the proposed CRLB-based collaborative RTT localization, which combines RTT ranging and fingerprinting methods using a CRLB-based outlier detection strategy. The offline process involves constructing the RTT fingerprint database and the CRLB map of the testing area. Initially, the testing area is divided into a set of grid points (GPs) with known coordinates. Using the coordinates of GPs and Wi-Fi APs, along with the ranging variance model (16), the CRLB map is generated (see Fig. 3). Constructing the RTT fingerprint database involves calculating the theoretical distance between GPs and Wi-Fi APs. A reference RTT fingerprint $\boldsymbol{\Xi}_i$ linked with the ith GP is expressed as

$$\boldsymbol{\Xi}_i = \left[d_{i1}, \ldots, d_{ij}, \ldots, d_{\mathrm{in}}, x_i, y_i\right], \quad j \in [1, n] \tag{17}$$

where (x_i, y_i) represents the coordinates of the ith GP, d_{ij} denotes the theoretical distance between the ith GP and the jth AP, and n denotes the number of Wi-Fi APs. The RTT fingerprint database is obtained by calculating the fingerprint data for all GPs.

The online phase includes RTT ranging error compensation, outlier detection, and RTT collaborative positioning. RTT ranging errors are compensated using a semiparametric error model [43], which describes errors (e.g., multipath and NLoS)

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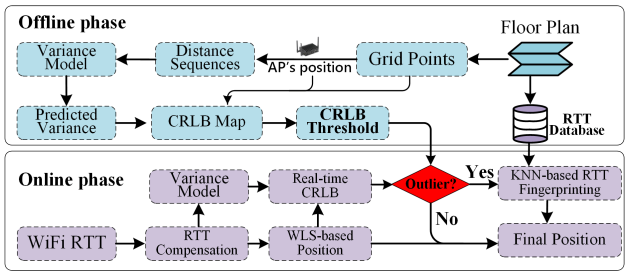

Fig. 4. Flowchart of the CRLB-based collaborative RTT ranging and fingerprinting approach.

as a sum of linear and nonlinear terms as follows:

$$\boldsymbol{e}_{n\times 1} = \boldsymbol{b}_{n\times 2}\boldsymbol{x}_{2\times 1} + \boldsymbol{S}_{n\times 1} + \Delta \tag{18}$$

where $\boldsymbol{e}$ is the ranging error matrix for n Wi-Fi APs, $\boldsymbol{bx}$ is the linear term of ranging errors, $\boldsymbol{x}$ is the parameter matrix, $\boldsymbol{b}$ is a nonsingular matrix, $\boldsymbol{S}$ denotes the nonlinear term, and Δ is the measurement noise. More details for the semiparametric model are provided in [43]. Following ranging error compensation, RTT ranging positioning is performed using the weighted LS (WLS) method, yielding a coarse-grained WLS-based position. Then, the CRLB-based outlier detection strategy evaluates this position.

Outlier detection identifies outliers by comparing real-time calculated CRLB values to predefined thresholds. Since CRLB indicates theoretical localization accuracy, a high CRLB represents a positioning outlier. In cases where small CRLB values occur in inaccessible areas, a low CRLB also signifies an outlier. Considering these cases, we defined the following conditions for outlier detection:

$$\eta > \delta_{\max} \| \eta < \delta_{\min} \tag{19}$$

where η, $\delta_{\max}$, and $\delta_{\min}$ are the real-time calculated CRLB and the maximal and minimal thresholds, respectively; and η is calculated using the WLS-based position, coordinates of Wi-Fi APs, and variance model (16). In this work, $\delta_{\max}$ and $\delta_{\min}$ are set to 1.2 and 0.9, respectively.

Collaborate RTT positioning is executed if (19) is met, indicating an outlier and leading to the rejection of the WLS-based position. The algorithm then eliminates the outlier and performs RTT-based fingerprinting using the KNN approach [21] and RTT fingerprint database. The outcomes of the KNN-based RTT fingerprinting serve as the estimated position if an outlier is identified. Otherwise, the WLS-based position is output as the final estimation. Fig. 4 presents the detailed process of the CRLB-based collaborative RTT positioning method.

B. EGAWR Positioning

In our previous work [25], we validate the feasibility of using magnetic positioning to improve the performance of Wi-Fi FTM localization in real-life scenarios. The primary process of the geomagnetic-aided Wi-Fi FTM positioning involves overlapping searching area (OSA) definition and coincident point searching. Based on the geometry relationship between the points within OSA and Wi-Fi APs, a set of coincident points can be obtained, and their average coordinates serve as the estimated fine-grained Wi-Fi position. However, the purpose of [25] is to improve the RTT localization accuracy by incorporating magnetic positioning. Optimization of the ranging positioning and online positioning results evaluation was not the focus of the study.

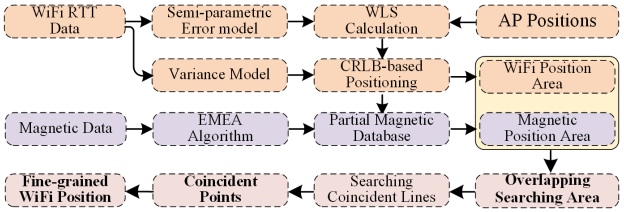

Fig. 5. Framework of the EGAWR positioning.

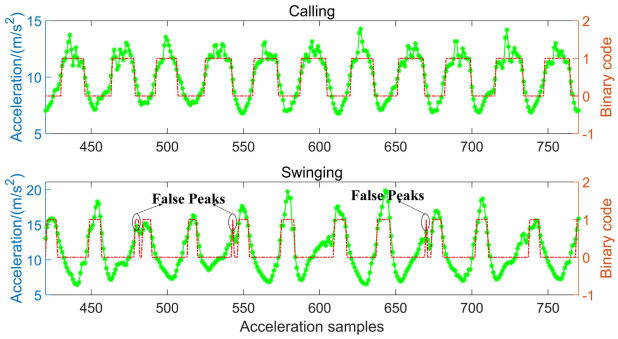

Fig. 6. Smoothed acceleration data and transformed binary codes under calling and swinging states.

As illustrated in Fig. 5, the EGAWR positioning primarily undergoes optimization through CRLB-based dynamic evaluation and ranging compensation. A more accurate Wi-Fi ranging positioning method provides a more reliable partial magnetic database for performing magnetic positioning, which in turn helps to define an OSA closer to the ground-truth position. In this work, the EGAWR positioning plays an important role in PF optimization. The coincident points obtained from the OSA contribute to the reconstruction of the degenerated particle swarm (described in Section IV-D), ensuring the reliability of particles after multiple resamplings.

C. Improved PDR Framework

1) Binarization-Based Step Detection: Fig. 6 shows the acceleration data of calling and swinging states. The data are measured with a sampling rate of 50 Hz. It can be seen that the waveforms of the binary codes resemble pulse signals, where each step corresponds to one pulse signal. The advantage of this method is that it does not require detecting peak or zero states of the walking process but only checks for the occurrence of a pulse signal. Fig. 6 illustrates that the pulse signal corresponding to the false peak is narrower than that of the normal walking process. Therefore, utilizing binarized codes for step detection is more robust against false peaks.

To detect the step, the approach only requires a binary threshold (which will be described in the following) and a sliding window with a size of S. As shown in Algorithm 1, the identification process can be described with three phases: acceleration binarization, false peak detection, and step detection and confirmation.

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Algorithm 1 Binarization-Based Step Detection

Input: acceleration data acc, binarization threshold t, slide window size S, etc.
Output: Step detection result

```
//Smooth data;
acc=smooth(acc,7)      // mean filter;
Acceleration binarization;
// Step detection;
a ←0, step ←0;
for i ← 1 to length(acc_bin) do
    if acc_bin(i) = 1 then
        a ← a + 1;
    else if a < S/2 && acc_bin(i) == 0 then
        a ← 0      // false peak;
    end
    else if a > S/2 && acc_bin(i) == 0 then
        a ← a + 1      // step confirmation;
    end
    if a > S then
        a ←0, step ← step + 1      // detected steps;
    end
end
```

First, the mean filter is applied to smooth the real-time measured acceleration data. After that, the smoothed acceleration is binarized according to the threshold. When all these works are completed, false peak detection starts. We set a temporary number a to count the binary code "1." a increases by one if the real-time measured accelerations are greater than the binarization threshold. If $a < (S/2)$, but there are the following codes "0," it can be concluded that a false peak occurs, and the detection work restarts. When $a > (S/2)$, which means that a step is ready to be identified, the algorithm continues to count code "0" to confirm that one step is detected. In the meantime, a continues to increase by one if code "0" is detected. Finally, the step detection is finished if the condition $a > S$ is satisfied. Generally, S is linked to the total number a_{sum} of acceleration data of one-step period (under walking forward state). Based on our analysis, we recommend to take the half value of a_{sum} as the window size. In this work, the number of S is 18.

We also design an iterative approach to find the threshold of binarization. The offline work needs to count the true number of steps with different phone-holding styles (e.g., horizontally, calling, or swinging). By inputting the measured acceleration data into Algorithm 2, a threshold for binarization can be automatically generated.

2) Adaptive Step Length Estimation: In the case of practical application, the nonlinear Weinberg model [40] is a widely applied method, which estimates stride size based on the maximal and minimal accelerations as

$$l = K \times \sqrt[4]{a_{\max} - a_{\min}} \tag{20}$$

where K is a scale factor that requires training during the offline stage. However, the pedestrian walking process is not constant, which limits (20) to estimate the step length using a

Algorithm 2 Iterative Method to Find Binary Threshold

Input: acceleration data acc, total number of steps N, boolean $flag$, etc.
Output: binary threshold t

```
flag ← true, t ← 9.80;
//Execution;
while flag do
    t = t + 0.1      // increment;
    if t < max(acc) then
        Calculate step number n using Algorithm 1
    end
    if abs(N − n) ≤ 1 then
        flag ← false
    end
end
```

constant scale factor. To this end, we consider the frequency of taking steps and the acceleration standard deviation of one step to assign dynamic values to K. The model is as follows:

$$l = \frac{\sqrt[5]{\sqrt{f/a_{\text{std}}}}}{2} \times \sqrt[4]{a_{\max} - a_{\min}} \tag{21}$$

where f and a_{std} denote the stride frequency and the standard deviation of acceleration, respectively. a_{std} is further given by

$$a_{\text{std}} = \sqrt{\frac{\left[(a_{\max} - \bar{a})^2 + \cdots + (a_{\min} - \bar{a})^2\right]}{h}} \tag{22}$$

where h is the number of values from $a_{\max}$ to $a_{\min}$ and $\bar{a}$ is the mean value of the h accelerations. Generally, the changing process of acceleration from the maximum to the minimum values corresponds to half period of one step. Based on the sampling rate H of the accelerometer, the stride frequency can be defined as

$$f = \frac{H}{2h}. \tag{23}$$

Different stride frequencies and acceleration standard deviations reflect varying walking speeds, which correspond to different stride sizes. Therefore, our model can adaptively adjust K to estimate step length and reduce the training work of K of the classic model during the offline phase.

3) Geomagnetic Waveform-Based Heading Optimization: The heading angle, also known as the azimuth, is the angle between the truth north and the walking direction. The Mahony complementary filter (MCF) [23] is a widely used method for heading estimation. It represents the carrier's attitudes using a quaternion vector, denoted as $\boldsymbol{q} = [q_0, q_1, q_2, q_3]^T$, and calculates the heading angle as

$$\Psi = \arctan \frac{2(q_1 q_2 + q_0 q_3)}{1 - 2\left(q_2^2 + q_3^2\right)}, \quad \Psi \in [0°, 360°) \tag{24}$$

where the values of $\boldsymbol{q}$ are updated using gyroscope, accelerometer, and magnetometer data. Further details on MCF can be found in [23], [34], and [36]. Generally, MCF accurately estimates azimuth when the phone is placed horizontally. However, errors may arise due to changes in the smartphone's

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orientation during walking. To address this, we propose a geomagnetic waveform-based heading angle optimization method, comprising three steps: geomagnetic sequence matching, coordinate inverse calculation, and outlier control.

Step 1 (Geomagnetic Sequence Matching): Based on [18] and [19], magnetic sequences exhibit a strong correlation with indoor topologies. To perform geomagnetic sequence matching, magnetic waveforms and actual indoor locations are mapped as

$$\boldsymbol{mg} = (x_1, y_1, B_1), \ldots, (x_i, y_i, B_i), \ldots, (x_n, y_n, B_n) \quad (25)$$

where $\boldsymbol{mg}$ is the mapped reference geomagnetic sequence; B_i denotes the ith magnetic field strength; and n and (x_i, y_i) represents the sequence length and the corresponding position of B_i, respectively. After measuring the magnetic data along the reference path from a known starting point to another position, the mapping process is executed using (25) by interpolating coordinates.

For online matching, a dynamic time warping (DTW) algorithm [19] matches the real-time measured magnetic sequence with the reference pattern [constructed by (25)] at a specific step size. In this work, the step size is the same as the sampling rate of the magnetometer. If the sampling frequency is N, each match begins by moving N data points forward on the reference pattern. The length of the data for online magnetic matching is $5N$. If the measured sequence matches a $5N$-length segment from the reference pattern, the position corresponding to index $5N$ of the segment is output.

Step 2 (Coordinate Inverse Calculation): As shown in Fig. 7, using two consecutive positions from Step 1, the theoretical azimuth between them can be calculated. In a geographic coordinate system (GCS), the azimuth of the line connecting two positions is determined by

$$\Psi_k = \arctan \frac{e_k - e_{k-1}}{n_k - n_{k-1}} = \arctan \frac{\Delta e}{\Delta n} \quad (26)$$

where Ψ_k is the azimuth connecting two consecutive positions at times $k-1$ and k; (n_{k-1}, e_{k-1}) and (n_k, e_k) are the coordinates of the two consecutive positions along the north and east axes in the GCS, respectively; and Δe and Δn denote the coordinate increments. Depending on the walking and north directions, the azimuth falls into different GCS quadrants (as shown in Fig. 7). The final azimuth Ψ_k at time k is calculated using the methods in Table I.

Step 3 (Outlier Control): Although the theoretical azimuth can be determined using (26) and the methods of Table I, magnetic mismatches can lead to outliers in the results. To control outliers, we use the coordinate increment in (26) and the two calculated consecutive azimuths as follows:

$$\Psi_k = \begin{cases} \arctan \dfrac{\Delta e}{\Delta n}, & (0 < d < \delta_d)\&(\Delta_\Psi < \delta_\Psi) \\ \arctan \dfrac{2(q_1 q_2 + q_0 q_3)}{1 - 2\left(q_2^2 + q_3^2\right)}, & \text{other} \end{cases} \quad (27)$$

where d is the plane distance between the two consecutive positions at times $k-1$ and k; $d = (\Delta e^2 + \Delta n^2)^{1/2}$; Δ_Ψ is the difference between the two consecutive azimuths; $\Delta_\Psi = |\Psi_k - \Psi_{k-1}|$; Ψ_k and Ψ_{k-1} are the azimuths at times k and $k-1$, respectively; and δ_d and δ_Ψ are the thresholds, set to $\delta_d = 2$ m and $\delta_\Psi = 90°$ in this work, respectively. Equation (27) indicates that the geomagnetic waveform-based heading optimization will be employed if no mismatch occurs. Otherwise, the heading estimated by MCF is used as the final heading angle.

It should be noted that heading optimization is not executed if the real-time measured magnetic sequence is shorter than $5N$. In such cases, the system relies on the heading estimated by the MCF. If the $5N$-length condition is met, the proposed optimization method is applied. The real-time measured magnetic data follow the first-in–first-out principle, which means that the first N data points are removed from the sequence after each match.

TABLE I
AZIMUTH CALCULATION BASED ON DIFFERENT QUADRANTS

Quadrants	Ranges	Δe, Δn	Calculation
I	$0° \sim 90°$	$\Delta e > 0, \Delta n > 0$	$arctan \frac{\Delta e}{\Delta n}$
II	$90° \sim 180°$	$\Delta e > 0, \Delta n < 0$	$arctan \frac{\Delta e}{\Delta n} + 180°$
III	$180° \sim 270°$	$\Delta e < 0, \Delta n < 0$	$arctan \frac{\Delta e}{\Delta n} + 180°$
IV	$270° \sim 360°$	$\Delta e < 0, \Delta n > 0$	$arctan \frac{\Delta e}{\Delta n} + 360°$

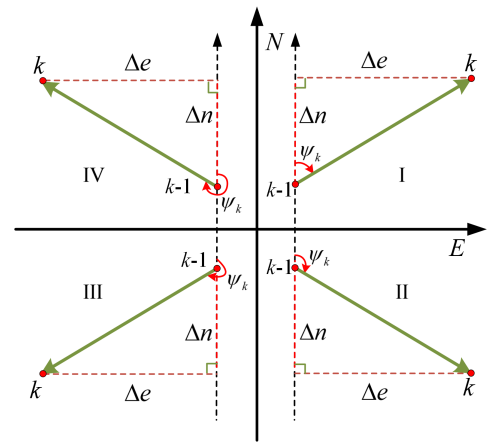

Fig. 7. Relationship between azimuth and quadrants of the GCS.

D. PF Optimization

The PF is a popular approach that infers a nonlinear filter problem [44]. Generally, PF always suffers from the sample impoverishment problem. Although adopting resampling strategies [45] can mitigate this issue, it still exists after multiple resamplings. To keep high-performance of the PF, we analyze that enhancing the reliability of the particle swarm and resampling credible samples are essential.

1) Particle Swarm Reconstruction Using Wi-Fi RTT: If a particle swarm with bias is selected, the resampling strategy cannot provide efficient information for the simulation. As a result, fewer particles are selected from the particle swarm due to the bias. To address this issue, we propose a particle swarm reconstruction strategy based on the geomagnetic-aided Wi-Fi RTT positioning. Referring to Fig. 5, a set of coincident points can be obtained after searching the overlapping area. Unlike randomly generated particles, the coincident points can

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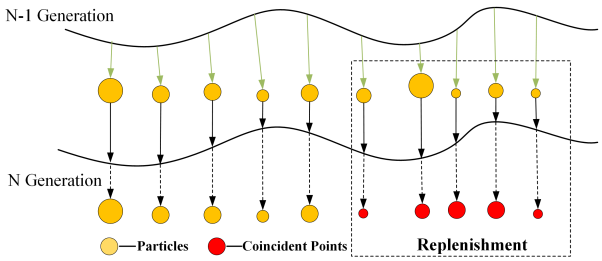

Fig. 8. Particle swarm reconstruction.

be regarded as the particles closest to the ground-truth position, representing the possible true posterior state of the system. Therefore, as shown in Fig. 8, we dynamically replenish the coincident points into the particle swarm during the filtering process. This operation ensures the reliability of particles and provides positive feedback for the resampling process.

2) Resampling With Map Information: In the case of a moving system in indoor environments, the particle swarm propagates with the system movement. In general, It is impossible for a pedestrian walking along the roads inside the building to pass through walls or enter inaccessible areas. Therefore, resampling should consider the geometric relationship between the particles' positions and indoor topologies based on the map information. To determine whether to remove particles, we introduce a coordinate constraint strategy as follows:

$$\begin{cases} x_k > x_{p\max} || x_k < x_{p\min} \\ y_k > y_{p\max} || y_k < y_{p\min} \end{cases} \quad \forall (x_k, y_k) \in \mathbb{R}^2 \tag{28}$$

where $(x_{p\min}, y_{p\min})$ and $(x_{p\max}, y_{p\max})$ represent the minimal and maximal coordinates of the map boundary, respectively, and (x_k, y_k) is the position of the particle. The advantage of map-aided resampling is that it brings the estimated results closer to the ground-truth position, providing positive feedback for position estimation and avoiding possible divergence.

E. Wi-Fi RTT/Magnetic/PDR Integration Based on IPF

1) System State and Observation Models: Our previous works in [23] and [36] have proven that integrating PDR with Wi-Fi or geomagnetic data using the EKF effectively controls the drifts in PDR locations. Therefore, we select the EKF-based frameworks [23], [36] to represent the system state of the Wi-Fi RTT/magnetic/PDR fusion model. When the system operates within an area where Wi-Fi is available, the system state model is defined as

$$\begin{cases} \boldsymbol{x}(k) = \boldsymbol{F} \bullet \boldsymbol{x}(k-1) + \boldsymbol{\Omega} \bullet \boldsymbol{U}(k-1) + \boldsymbol{w}(k-1) \\ \boldsymbol{z}(k) = \boldsymbol{H} \bullet \boldsymbol{g}(k) + \boldsymbol{M} \bullet \boldsymbol{S}(k) + \boldsymbol{q}(k) \end{cases} \tag{29}$$

where $\boldsymbol{x}(k)$ and $\boldsymbol{x}(k-1)$ represent the PDR positions at time k and $k-1$, respectively; $\boldsymbol{U}(k-1)$ denotes the position given by Wi-Fi RTT positioning; $\boldsymbol{z}(k)$ is the filtered position; $\boldsymbol{S}(k)$ is a position that is near $\boldsymbol{x}(k)$; $\boldsymbol{F}$ and $\boldsymbol{\Omega}$ are the transition and measurement matrices, respectively; $\boldsymbol{H}$ and $\boldsymbol{M}$ are control matrices; $\boldsymbol{M} = -\boldsymbol{H}$; and $\boldsymbol{w}$ and $\boldsymbol{q}$ describe the system and measurement noise, respectively. More details for the definitions of $\boldsymbol{F}$, $\boldsymbol{\Omega}$, $\boldsymbol{H}$, and the RTT/PDR fusion are referred to our work [36].

When the Wi-Fi RTT/magnetic/PDR system enters an area where Wi-Fi is unavailable, the system switches to the EKF-based magnetic/PDR fusion as the system state

$$\begin{cases} \boldsymbol{x}(k) = \boldsymbol{F} \bullet \boldsymbol{x}(k-1) + \boldsymbol{w}(k-1) \\ \boldsymbol{z}(k) = \boldsymbol{H} \bullet \boldsymbol{g}(k) + \boldsymbol{q}(k) \end{cases} \tag{30}$$

where $\boldsymbol{x}(k)$, $\boldsymbol{x}(k-1)$, $\boldsymbol{z}(k)$, $\boldsymbol{w}$, and $\boldsymbol{q}$ represent the same meanings of the parameters in (29); $\boldsymbol{g}(k)$ is the geomagnetic positioning result; and $\boldsymbol{F}$, $\boldsymbol{H}$, and the magnetic/PDR fusion process are further given in our work [23].

Equations (29) and (30) represent the kinematic model of the proposed system. By executing a classic Kalman filtering process, the state of the proposed system can be obtained. In general, the proposed Wi-Fi RTT/magnetic/PDR prefers to use (29) as the system state when all positioning signals are available, and it will switch to (30) if Wi-Fi is unavailable.

The system observations of the fusion model consist of three components: geomagnetic data $\boldsymbol{G}$, RTT ranging data $\boldsymbol{d}$, and PDR information (step length l and heading angle Ψ)

$$\text{obser} = [\boldsymbol{G}, \boldsymbol{d}, l, \Psi]^T \tag{31}$$

where $\boldsymbol{G}$ and $\boldsymbol{d}$ are expressed as

$$\boldsymbol{G} = \left[B_x, B_y, B_z, B_h, B\right]^T \tag{32}$$

$$\boldsymbol{d} = [d_1, \ldots, d_i, \ldots, d_n]^T \tag{33}$$

where B_x, B_y, and B_z, B_h, and B represent the three-axis components, the horizontal, and the total intensity of the magnetic field, respectively; and d_i is the RTT ranging data from the ith Wi-Fi AP.

2) Fusion Model Construction: The detailed fusion process is described as follows.

Step 1 (Initialization): The fusion system first loads the magnetic database, map information, and CRLB threshold for dynamic evaluation. A particle swarm with N samples is generated within the target area, with the attributes as

$$\boldsymbol{pset}_i = \left\{\boldsymbol{G}_i, \boldsymbol{d}_i, \tilde{l}_i, \tilde{\Psi}_i, w_i, x_i, y_i\right\} \tag{34}$$

where w_i and (x_i, y_i) are the weight and coordinates of the ith particle, respectively. The initial weight is set to $1/N$ and the coordinates of the particles are obtained by randomly selecting positions within the area. To ensure the particles searching in all directions, $\tilde{l}_i$ and $\tilde{\Psi}_i$ are assigned by

$$\begin{cases} \tilde{l}_i = l_i + \epsilon_l, \tilde{\Psi}_i = \Psi_i + \epsilon_\Psi, & \forall r > 0.5 \\ \tilde{l}_i = l_i - \epsilon_l, \tilde{\Psi}_i = \Psi_i - \epsilon_\Psi, & \forall r < 0.5 \end{cases} \tag{35}$$

where ϵ_l and ϵ_Ψ are the assigned Gaussian errors of the step length and heading angles, $\epsilon_l \sim \mathcal{N}(0, \delta_l^2)$ and $\epsilon_\Psi \sim \mathcal{N}(0, \delta_\Psi^2)$, respectively; and r is a random number.

It should be noted that the above process will be executed if the system is initialized within the Wi-Fi area. In cases where the system starts without Wi-Fi signals, the EKF-based magnetic/PDR is executed to determine positions. Once the system reenters an area with Wi-Fi signals, the proposed Wi-Fi RTT/magnetic/PDR system will be activated again. The initialization process is completed upon inputting PDR information.

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Step 2 (Positioning Calculation): After initialization, the Wi-Fi, magnetic, and PDR localization modules generate results based on the observations. We utilize the first positioning result of the CRLB-based Wi-Fi RTT positioning as the initial position of the fusion system. If no Wi-Fi signals are observed, the Wi-Fi positioning module is not executed, and magnetic positioning results are used as the initial system position.

Step 3 (System State Prediction and Particle Weight Update): Based on the positioning results in Step 2, if Wi-Fi positioning is available, the system state is predicted using (29); otherwise, (30) is used. The weights of particles are updated by

$$w_i = \frac{1}{\sqrt{2\pi\delta}} \exp\left(-\frac{d_i^2}{2\delta^2}\right) \tag{36}$$

where δ represents the system variance and d_i denotes the distance between the ith particle's position and system state. The weights are normalized after updating all particles' weights according to (36).

Step 4 (Map-Aided Resampling): In this step, the samples that are outside the map or enter inaccessible areas will be removed from the particle swarm. The remained samples are evaluated by using the polynomial resampling strategy [45]. The greater the weight of the particle, the greater the probability of being selected and copied. It should be noted that whether to execute Step 4 depends on whether Wi-Fi positioning is available. If not, the system skips Step 4 and goes to Step 5.

Step 5 (Fusion Position Estimation): If Step 4 is executed, the fusion position is output by computing the average coordinates of the resampled particles

$$\begin{cases} X_k = \dfrac{1}{m}\displaystyle\sum_{i=1}^{m} x_k \\ Y_k = \dfrac{1}{m}\displaystyle\sum_{i=1}^{m} y_k \end{cases} \tag{37}$$

where m denotes the number of resampled particles. If no resampled particles are obtained from Step 4, fusion results of the EKF-based magnetic/PDR system are output as the final result (details can be found in Fig. 1).

Step 6 (Particle Swarm Reconstruction): To improve the diversity and reliability, we utilize the coincident points obtained from the geomagnetic-aided Wi-Fi positioning module to replenish the particle swarm. If there are m coincident points, the replenishment is finished by replacing the coordinates of the last m samples of the particle swarm (as shown in Fig. 8).

The detailed fusion process is illustrated in Algorithm 3. If the system starts working in an area where no Wi-Fi signals, the current system is executed by the EKF-based magnetic/PDR fusion until switching to the Wi-Fi RTT/magnetic/PDR fusion when Wi-Fi signals are input. This mechanism ensures localization continuity when transitioning between areas with and without Wi-Fi signals.

Algorithm 3 Wi-Fi RTT/Magnetic/PDR Fusion

Input: particle number N, RTT data $\hat{\boldsymbol{d}}_k$, error control terms K_p and K_I, map data, magnetic and RTT database, KF parameters $\boldsymbol{F}$, $\boldsymbol{H}$, $\boldsymbol{\Omega}$, $\boldsymbol{w}$, and $\boldsymbol{q}$, CRLB threshold $crlb$, etc.

Output: fusion positioning result (X, Y)

```
//Initialization;
Load map information and fingerprint database;
Initialize N particles within the testing area;
Sensors data preprocessing;
//Fusion;
while measurements received & step detected do
    Calculate heading Ψ and step length L;
    if d̂_k is available then
        Encode particles with (34);
        Compensate d̂_k using semiparametric model;
        Perform WLS-based WiFi positioning and predict
        ranging variances;
        Calculate real-time CRLB;
        if CRLB > crlb then
            Perform RTT fingerprinting      // outlier;
        else
            Output WLS-based position;
        end
        Perform enhanced geomagnetic-aided WiFi RTT
        positioning module;
        Obtain the coincident points set;
        Calculate PDR position;
        Predict system state using (29);
        Particles weights update;
        Resampling;
        Fusion results calculation;
        Particle swarm reconstruction;
    else
        Calculate PDR position;
        Perform magnetic positioning;
        Make magnetic/PDR fusion based on (30);
        Fusion results calculation by executing EKF;
    end
end
```

V. Experiments and Results

A. Experimental Setup

Experiments were conducted on the second floor of a typical working scenario (as shown in Fig. 9), with eight Compulab Wi-Fi access points (APs) installed 5.2 m above ground level, resulting in an NLoS condition. We selected 188 reference points at 1.2-m intervals and used a total station for precise coordinates to construct the magnetic and RTT databases. Magnetic data collected at these points resulted in a database of 1756 interpolated GPs at 0.3-m intervals. The RTT database was constructed as described in Section IV-A.

We utilize a Google Pixel 3 phone for testing, with Wi-Fi RTT ranging and sensor sampling rates of 5–50 Hz, respectively. The IPF was configured with 2000 particles. To simulate

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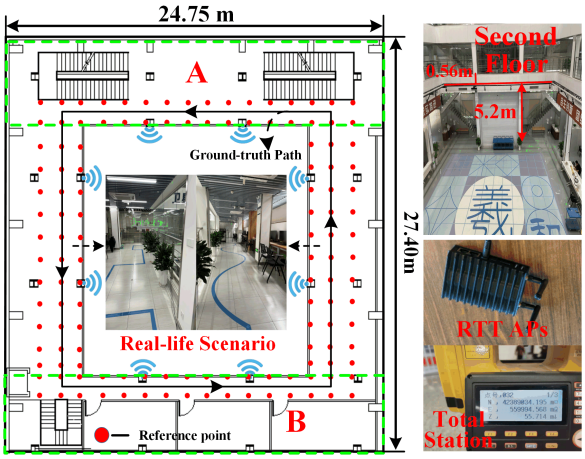

Fig. 9. Floor plan and real-life scenario of the experimental area and the testing device.

scenarios without Wi-Fi signals, areas A and B (within green lines in Fig. 9) were assumed to have no signal. The tester completed three cycles along the reference path, recording Wi-Fi, magnetic, and PDR data. The testing points are selected every 2.4 m and the corresponding ground-truth coordinates are measured by using the total station. The testing data are analyzed on a laptop with 16-GB RAM and 2.3-GHz CPU.

B. CRLB-Based Wi-Fi RTT Positioning Evaluation

1) RTT Collaborative Positioning Evaluation: We first investigate the proposed CRLB-based Wi-Fi RTT localization method against the KNN-based RTT fingerprinting method [9], the RSS-based distance and FTM ranging fusion (RSS-FTM) positioning [27], and the LSs method [34]. We utilize the meaning positioning error, root-mean-square error (RMSE), and 75th and 90th percentile errors to make discussions. The testing results and CDFs of different methods are presented in Table II and Fig. 10, respectively.

Table II reveals that the localization error and RMSE of LS-based RTT positioning are 3.77 and 4.37 m, respectively. While RSS-FTM incorporates RSS-based distance to enhance positioning accuracy, it only yields a marginal improvement of 0.52 m. Conversely, KNN-based RTT fingerprinting outperforms the ranging-based methods in terms of mean accuracy and RMSE within our testing site. Based on this, the proposed CRLB-based collaborative RTT positioning achieves an accuracy of 2.17 m, outperforming LS, RSS-FTM, and RTT fingerprinting. The same conclusions can be made by comparing the RMSE, and 75th and 90th percentile errors. Fig. 10 shows that the proposed RTT positioning method consistently performs better than LS, RSS-FTM, and RTT fingerprinting across various confidence levels. The testing results indicate that the proposed CRLB-based dynamic evaluation can effectively eliminate outliers and enhance accuracy.

2) Geomagnetic-Aided Wi-Fi RTT Positioning Evaluation: According to Table II and Fig. 10, we also compare the performance of the proposed EGAWR positioning with the classic geomagnetic-aided Wi-Fi RTT positioning (GAWR) [25] and the RSS-FTM method [27]. Based on the dependable collaborative RTT positioning results, EGAWR achieves a mean precision of 1.54 m, being an improvement of 0.28 m compared to GAWR. EGAWR reduces the RMSE and 75th percentile error of GAWR by 0.1 and 0.34 m, respectively. The proposed EGAWR also performs better than RSS-FTM, reducing the mean positioning error and RMSE by 1.71 and 1.76 m. Moreover, Table II indicates the favorable performance of EGAWR compared to the other five approaches, as further demonstrated by the cdf comparisons in Fig. 10. Therefore, it can be concluded that a more accurate RTT positioning method enhances the precision of geomagnetic-aided Wi-Fi RTT positioning. These promising results offer a reliable supplement for particle reconstruction, as stated in Section IV-D.

TABLE II
POSITIONING ERRORS OF DIFFERENT WI-FI METHODS

Methods	Mean/(m)	RMSE/(m)	75th/(m)	90th/(m)
CRLB-based RTT Positioning	2.17	2.73	3.30	4.17
RTT-based Fingerprinting [9]	2.60	3.05	3.63	4.49
Least Squares [34]	3.77	4.37	5.09	6.59
GAWR [25]	1.82	2.08	2.51	3.20
RSS-FTM [27]	3.25	3.74	4.17	5.00
EGAWR	1.54	1.98	2.17	3.27

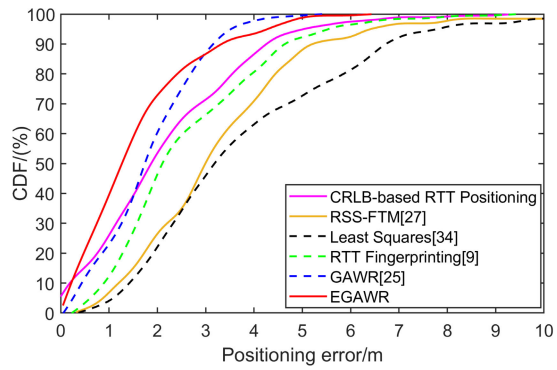

Fig. 10. CDF comparisons of different Wi-Fi positioning methods.

C. Improved PDR Framework Evaluation

1) Step Detection: We undertake a comparison of our step detection algorithm with the well-known FSM method [36] and the peak detection algorithm [38]. Four walking states are investigated: walking forward while holding phones horizontally (WFH), walking backward while holding phones horizontally (WBH), walking forward while listening phones (WFL), and walking forward while swinging phones (WFS). We utilize the over(+)/miss(−) detection rates to evaluate performance.

Table III shows that the three methods obtain high-precision step detection results during the WFH state. The results of the peak detection method fluctuate greatly, with a miss-detection rate of −17.3% and an overdetection rate of +10% in the WFL and WFS states, respectively. This indicates that false peaks in the WFS and WFL states seriously affect the robustness of the peak detection algorithm. The FSM method performs better than peak detection but still delivers miss-detection rates of −2.0% to −5.3% under WFL and WFS states. The

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TABLE III
STEP DETECTION RESULTS OF DIFFERENT METHODS

States	Number of Steps	FSM [36]	Peak Detection [38]	Proposed
WFH	150	152(-0.7%)	152(+1.3%)	150(+0.0%)
WFL	150	153(-2.0%)	124(-17.3%)	149(-0.7%)
WFS	150	142(-5.3%)	165(+10.0%)	152(+1.3%)
WBH	100	101(+1.0%)	101(+1.0%)	101(+1.0%)

TABLE IV
STEP LENGTH ESTIMATION RESULTS OF DIFFERENT METHODS

Stride Frequency	Test Number	Shin [39]	Weinberg [40]	Proposed
Low	First	78.12	59.19	62.75
(1.0m/s	Second	70.57	61.93	64.16
∼1.1m/s)	Error	21.29/16.7%	5.68/4.5%	1.39/1.1%
Normal	First	75.77	62.81	64.65
(1.2m/s	Second	70.17	61.97	63.85
∼1.3m/s)	Error	18.54/14.6%	2.62/2.1%	1.1/0.86%
Fast	First	66.88	59.84	60.56
(1.4m/s	Second	60.86	59.69	61.37
∼1.5m/s)	Error	6.02/4.7%	7.87/6.2%	5.38/4.2%
Summation of error		45.85/12%	16.17/4.2%	7.87/2.1%

Note: the unit of walking distance in this table is *m*.

proposed method works well across four walking states, with an error rate of less than 2%. This is thanks to the fact that binarization-based step detection no longer depends on the change of acceleration to detect steps, thus mitigating the influence of false peaks on detection outcomes.

2) Step Length Estimation: We compare the proposed step length method with well-known models such as Shin model [39] and Weinberg model [40]. The tester walked along a straight line two times at varying stride frequencies: low, normal, and fast. The total length is 127.4 m. The absolute cumulative error rate (ACER) is defined to evaluate the models. The calculation method of ACER is defined as $(|\mathrm{SL} - L|/L) \times 100\%$, where SL and L are the estimated walking distance by the three methods and the ground-truth distance, respectively.

Table IV reveals that under three different walking speeds; the Shin model obtains ACERs of 16.7%, 14.6%, and 4.7%; and the Weinberg model achieves ACERs of 4.5%, 2.1%, and 6.2%, which are higher than those of the proposed estimation model. As a summation, our model delivers an ACER of 2.1%, while the ACERs of the Shin and Weinberg models are 12% and 4.2%, which are almost six times and two times higher than that of our model. Testing results demonstrate that utilizing predefined parameters cannot adapt well to different walking speeds, which in turn leads to estimation errors. However, the proposed adaptive step length model introduces stride frequency and the standard deviation of acceleration into the step length estimation, ensuring accurate estimations.

3) Heading Estimation: To test the performance of heading estimation methods, we compare the proposed heading optimization method with the MCF [23] and the magnetic compass-based approach [46]. The results from these methods are compared with standard heading, as shown in Table V.

TABLE V
HEADING ESTIMATION RESULTS OF DIFFERENT METHODS

Methods	Mean/(°)	RMSE/(°)	Max/(°)	90%Error/(°)
Proposed	4.26	6.10	36.24	9.97
MCF [23]	7.07	9.75	36.94	14.53
Compass [46]	23.09	28.91	67.15	53.81

TABLE VI
POSITIONING ERRORS OF METHODS WITH WI-FI IN AREAS A AND B

Methods	Mean/(m)	RMSE/(m)	75th/(m)	90th/(m)
PDR	1.25	1.38	1.59	2.06
Magnetic	1.93	2.40	2.67	3.97
WiFi RTT	2.18	2.74	3.28	4.18
Magnetic/WiFi	1.56	2.10	2.17	3.25
Magnetic/PDR [23]	0.98	1.12	1.29	1.80
WiFi/PDR [36]	0.81	0.93	1.14	1.45
Proposed	0.64	0.78	0.91	1.25

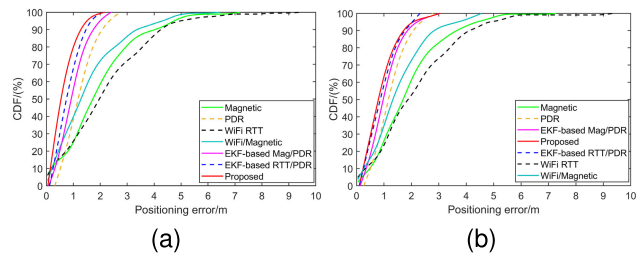

Fig. 11. CDF comparisons of different methods under different conditions. (a) Wi-Fi is available. (b) Wi-Fi is unavailable in areas A and B.

Table V shows that using the MCF for heading estimation achieves a mean accuracy of 7.07° and an RMSE of 9.75°, indicating good performance. However, the proposed heading optimization method improves these metrics to a mean error of 4.26° and an RMSE of 6.10°, being enhancements of 2.81° and 3.65° over the MCF method, respectively. In addition, the proposed method reduces the 90th percentile error of the MCF method by 6.56°. The testing results also reveal that although a magnetic compass can be used for heading estimation, its accuracy is limited, with an average accuracy of 23.09° and an RMSE of 28.91°. The 90th percentile error of the compass-based method exceeds 50°, which is almost 6× and 3.5× that of the proposed method and MCF, respectively. Therefore, it can be concluded that the proposed geomagnetic waveform-based heading optimization method can effectively improve heading estimation accuracy.

D. Fusion Framework Localization Performance

1) Wi-Fi Is Available in Areas A and B: We compare the fusion model with PDR, magnetic positioning, Wi-Fi RTT positioning, magnetic/Wi-Fi fusion, magnetic/PDR fusion [23], and Wi-Fi/PDR fusion [36]. The mean results of the ten runs of the system are presented in Table VI and Fig. 11(a).

The proposed system achieves a mean precision of 0.64 m, which is improved by 0.17–0.34 m compared to the Wi-Fi/PDR and magnetic/PDR fusion methods. The RMSE of the proposed system is 0.78 m, showing a reduction of 0.15–0.34 m compared to the Wi-Fi/PDR and magnetic/PDR

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TABLE VII
POSITIONING ERRORS OF METHODS WITHOUT WI-FI IN AREAS A AND B

Methods	Mean/(m)	RMSE/(m)	75th/(m)	90th/(m)
PDR	1.24	1.37	1.61	2.00
Magnetic	1.88	2.27	2.47	3.64
WiFi RTT	2.13	2.65	3.05	3.97
Magnetic/WiFi	1.82	1.78	1.61	2.91
Magnetic/PDR [23]	1.06	1.21	1.38	1.84
WiFi/PDR [36]	0.93	1.08	1.27	1.83
Proposed	0.91	1.09	1.23	1.69

fusion methods. The combination of PDR and magnetic positioning yields a precision of 0.98 m, which is improved by 0.95–0.27 m compared to magnetic positioning and PDR individually, respectively. Similarly, Wi-Fi RTT/PDR integration delivers an accuracy of 0.81 m, which is 1.37–0.44 m better than Wi-Fi RTT positioning and PDR, respectively. The cdf comparisons in Fig. 11(a) provide additional evidence of the superior performance of the proposed system across various percentile errors. These testing results confirm the effectiveness of the proposed Wi-Fi RTT/PDR/magnetic fusion system.

2) Wi-Fi Is Unavailable in Areas A and B: To evaluate the continuous localization capability of the proposed system, we simulate the system operating in a mixed environment where Wi-Fi signals are unavailable in areas A and B (see Fig. 9). The testing results are summarized in Table VII and Fig. 11(b).

Table VII demonstrates that even in areas without Wi-Fi signals, the proposed system achieves a mean accuracy of 0.91 m and an RMSE of 1.09 m, providing reliable meter-level continuous localization performance. The 75th and 90th percentile errors of the proposed system are 1.23–1.69 m. Fig. 11(b) further demonstrates the advantages of the proposed system across different percentile errors. Compared to the ubiquitous EKF-based magnetic/PDR fusion, the proposed system reduces the mean positioning error and RMSE by 0.15–0.12 m, respectively. These testing results demonstrate the localization capability of the proposed fusion system in scenarios where there are no Wi-Fi signals or the Wi-Fi module does not work.

3) Time Complexity of the Fusion System: Fig. 12 shows that with 500 particles, the mean error of the fusion system is approximately 0.9 m. As the number of particles increases, the mean positioning errors gradually decrease. When the number of particles exceeds 2000, the mean localization accuracy stabilizes around 0.65 m. Although the average execution time (AET) of each localization request grows as particles increase, the system still delivers results within less than 0.1 s when using 5000 particles. This indicates that the fusion framework exhibits low time complexity, making it suitable for long-term deployment on smartphones while achieving meter-level positioning accuracy.

E. Fusion Framework Localization Performance With Ranging Error Attack

We introduce errors to the ranging data by randomly selecting Wi-Fi APs. These errors follow a Gaussian distribution

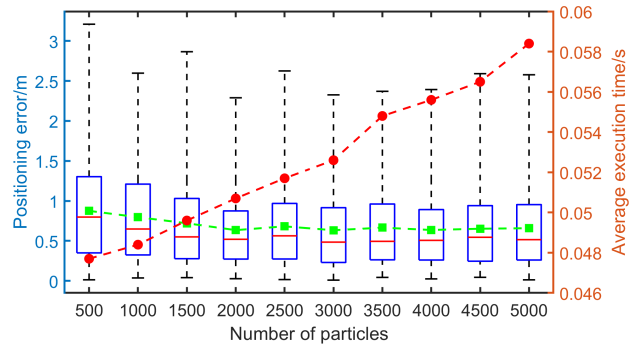

Fig. 12. Fusion positioning results and AET of every localization request when using different numbers of particles (mean accuracy is represented by the green dashed line).

TABLE VIII
POSITIONING ERRORS OF THE PROPOSED FUSION SYSTEM WITH RTT RANGING ERROR ATTACK

Number of WiFi APs	Mean/(m)	RMSE/(m)	50th/(m)	75th/(m)
1	0.67	0.89	0.98	1.46
2	0.69	0.88	0.95	1.31
3	0.73	0.89	1.03	1.51
4	0.75	0.95	1.08	1.58
5	0.78	0.95	1.16	1.47
6	0.80	0.99	1.16	1.66
7	0.85	1.09	1.18	1.68

with a mean value of zero and a variance of 5 m. Errors are assigned when a random number s is greater than 0.5, $s > 0.5, s \in [0, 1]$.

Table VIII illustrates that the mean localization error and RMSE of the system gradually increase as the number of selected APs increases. The mean error fluctuates between 0.67 and 0.85 m, and the RMSE fluctuates between 0.88 and 1.09 m. This testing result demonstrates that assigning errors to RTT range data can influence positioning performance. The CDF comparisons in Fig. 13 show that the localization performance is positively correlated with the number of selected APs. More selected APs result in more errors in the RTT observations, leading to worse positioning performance.

However, evaluating the system's error tolerance with an accuracy fluctuation of no more than 0.1 m yields interesting conclusions. Comparing the results from Tables VI and VII, when selecting two APs, the accuracy fluctuation is only a 0.05-m decrease. Moreover, the mean error increases by 0.09 m when three APs are selected. Therefore, we can conclude that the proposed system is error-tolerant (condition: mean accuracy fluctuation within 0.1 m) when the number of selected APs is fewer than 3. In real-life scenarios, it is common for one or two Wi-Fi APs to be temporarily blocked by pedestrians or moving devices. Our system still delivers a mean accuracy of about 0.85 m even in cases where all the measured data from APs contain additional errors.

F. Comparison With State-of-the-Art Approaches

We compared the proposed method with the KF-based RTT/PDR fusion system [12], the IPF-based magnetic/PDR fusion [24], and the enhanced PF (EPF)-based Wi-Fi

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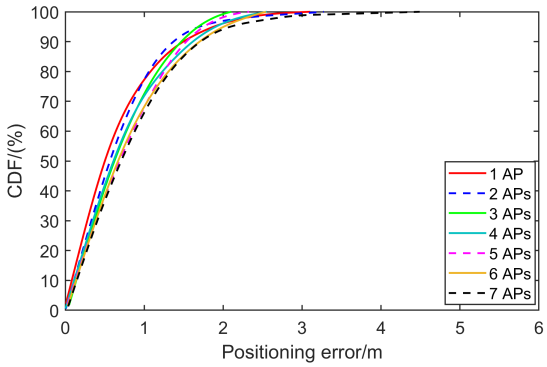

Fig. 13. CDF comparisons of the positioning results of the proposed system with additional ranging errors.

TABLE IX
POSITIONING ERRORS OF DIFFERENT FUSION METHODS

Methods	Mean/(m)	RMSE/(m)	75th/(m)	90th/(m)
Proposed	0.64	0.78	0.91	1.25
KF-based RTT/PDR fusion [12]	1.16	1.26	1.44	1.85
IPF-based Mag/PDR fusion [24]	1.00	1.11	1.31	1.71
EPF-based RTT/PDR fusion [34]	0.96	1.08	1.23	1.73

RTT/PDR fusion [34]. The results are presented in Fig. 14 and Table IX. Specifically, Fig. 14 is generated by plotting the estimated coordinates of the testing points, obtained through each fusion method, along with their corresponding positioning errors.

Table IX shows that the EPF-based RTT/PDR fusion and IPF-based magnetic/PDR fusion achieve localization accuracy of 0.96–1 m, with RMSEs of 1.08–1.11 m. The KF-based Wi-Fi RTT/PDR integration yields a mean error and an RMSE of 1.16–1.26 m respectively. In comparison with the three systems, the proposed system improves the mean accuracy by 0.32, 0.36, and 0.52 m and reduces the RMSE by 0.30, 0.33, and 0.48 m. Table IX indicates that the proposed system outperforms the compared state-of-the-art approaches regarding the 75th and 90th percentile errors. Fig. 14 reveals that the positioning results of the proposed system are closer to the ground-truth trajectory than the other three systems. These results demonstrate the advantages of the proposed system over state-of-the-art fusion methods in terms of positioning accuracy and stability.

G. Discussion

The application of smartphone-based indoor localization is constrained by factors such as accuracy, robustness, power consumption, and positioning investment. Table X shows that Wi-Fi RTT/RSS/PDR fusion [10] and Wi-Fi RTT/PDR fusion [12], [34], [35] have demonstrated good performance in reported testing sites. However, Wi-Fi/PDR systems still face challenges such as limited signal coverage (cannot work without Wi-Fi) and high positioning investment in large-scale environments, requiring more Wi-Fi device deployment. To mitigate investment costs, magnetic field and PDR integration [20], [24] have been explored, but magnetic data

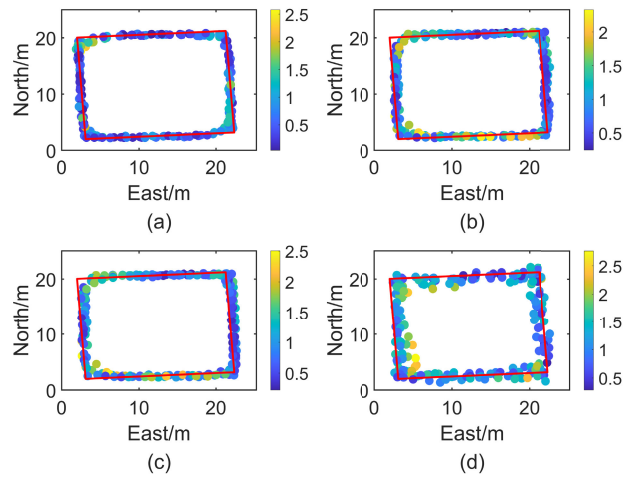

Fig. 14. Errors of the output positions of different fusion systems. (a) Proposed system. (b) EPF-based RTT/PDR fusion. (c) IPF-based Mag/PDR fusion. (d) KF-based RTT/PDR fusion. The red lines are the ground-truth trajectory and the color points represent the position errors with units of m.

TABLE X
COMPARISON OF THE DISCUSSED FUSION METHODS

Methods	RTT Function	Positioning without WiFi	Testing site	Accuracy /(m)
Proposed	✓	✓	Corridor	0.64
WiFi,PDR [10]	✓	✗	Office	0.57
WiFi,PDR [12]	✓	✗	Corridor	1.16
Mag,PDR [20]	✗	✓	Lobby environment	1.29
Mag,PDR [24]	✗	✓	Corridor	1.00
WiFi,PDR [34]	✓	✗	Corridor	0.96
WiFi,PDR [35]	✓	✗	Corridor	1.12
WiFi,PDR,Mag [47]	✗	✗	Office building	5.40

Note: The accuracy reported for the methods refers to 2D accuracy.

fluctuation [18] can affect system stability and accuracy. Our work addresses localization cost and accuracy by proposing a magnetic/PDR/Wi-Fi RTT fusion system. When the system operates outside the coverage of Wi-Fi signals, it employs geomagnetic and PDR to ensure localization continuity and accuracy. Therefore, there is no need for additional investment in Wi-Fi infrastructure. Although a magnetic/Wi-Fi/PDR system is developed in [47], this system is essentially still a Wi-Fi/PDR fusion positioning framework, optimized by a corner detection algorithm based on magnetic sequences. In addition, the Wi-Fi module in our system supports RTT ranging, which distinguishes our approach from that in [47]. In cases of power outages that render Wi-Fi infrastructure unavailable, our system can still rely on magnetic and PDR data to perform fusion positioning. However, the systems in [10], [12], [34], [35], and [47] will not be able to maintain positioning under such conditions.

For mobile phone deployment, power consumption and algorithm complexity are critical considerations. Despite using an IPF for fusion positioning, our system completes a localization request within 0.1 s (with 5000 particles), outperforming the EPF-based Wi-Fi RTT/PDR fusion (more than 0.15 s with 5000 particles) [34]. Furthermore, the utilization of geomagnetic, PDR, and Wi-Fi RTT positioning can be achieved through the sensors embedded within smartphones. Therefore, the integration of these three methods on smartphones is not

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constrained by the need for specialized hardware. Based on testing results and analysis, we conclude that integrating Wi-Fi RTT/magnetic/PDR using a PF for high-precision localization is feasible and efficient. The system's continuous localization capability in Wi-Fi-deprived scenarios, low positioning costs, and time complexity make it suitable for large-scale smartphone-based positioning.

VI. Conclusion

In this work, we have proposed a Wi-Fi RTT/magnetic/PDR fusion system based on an IPF and proved the good capability of the system. Under different localization scenarios where the Wi-Fi signal is available and unavailable, the proposed system obtains a mean accuracy of 0.64–0.91 m, and every fusion calculation can be finished within 0.1 s. In the future, we will introduce Wi-Fi FTM fingerprinting (including RSS and RTT) into the fusion system to enhance accuracy. Moreover, we will explore the 3-D fusion framework through the current version of the Wi-Fi/magnetic/PDR fusion system in more diverse testing environments, rather than being operated in the corridor environment.

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